Causality with Observational Data

MKT 566 · Fall 2026

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Before We Start

Housekeeping

- Next Tuesday, Nov. 17: in-class **causality exercise**, the methods from this week on simulated coupon data. Bring your laptop; R users, install the `MatchIt` package before class
- **Homework 4** due **Nov. 24** on Brightspace
- Thursday, Nov. 19: class time to **work on your group project**
- Final presentations: **Dec. 1 and Dec. 3**

Tuesday in one slide

- The effect of an action is **what happened minus what would have happened without it**. The second half, the **counterfactual**, is never observed
- **Randomization** solves it: a coin flip creates a control group that is the same kind of people as the treated group, so its average stands in for the missing world
- In advertising, **attribution** allocates credit and **incrementality** measures cause; only the second needs an experiment, and the two can disagree by a lot

> Today: what to do when **nobody flipped a coin**. The treatment happened, the data exist, and you still need the causal effect.

When You Cannot Randomize

Experiments are not always an option

- You cannot randomize a **city-wide policy**, a **law**, or a **platform-wide launch**: everyone gets it at once
- You cannot randomize **prices across customers** without fairness and legal trouble
- Sellers decide for themselves whether to **join a program**; customers decide whether to **redeem a coupon**. Nobody assigned them
- The decision was made **last year**, and the question arrives now. The data exist; the experiment does not

> **Observational data**: the treatment happened, but not because of a coin flip. Somebody, or something, chose who got it.

Experiments vs. observational data

	Experiment (RCT)	Observational data
Who assigns the treatment	you, at random	the market, the firm, the customers themselves
Causal inference	clean: compare the group averages	needs **assumptions** about why some units were treated
Cost	expensive, sometimes impossible	cheap, abundant, covers settings you could never experiment on

> The whole game with observational data is to **rebuild the control group** that randomization would have given you, and to be honest about the assumptions that takes.

The threat: confounding

A **confounder** is a variable that does two things at once:

1. it affects **who gets the treatment**
2. it affects **the outcome**

Example: stores that run **promotions** sell more than stores that do not.

Does the promotion cause the extra sales? What else could produce this pattern?

The threat: confounding, explained

- Maybe the promotion works. Or maybe store managers run promotions **when demand is already high** (a holiday, a local event, a good neighborhood)
- Then **demand** drives both: demand → promotion, and demand → sales. The promotion gets credit for sales that demand produced

> A confounder makes the treated and untreated groups **different kinds of units** before the treatment does anything. Comparing their averages mixes the treatment with the difference.

The selection problem

The customers, products, or stores that receive your marketing action are **not a random sample**:

- People who **see an ad** are more active shoppers (the platform targets them)
- Customers who **redeem a coupon** are more price sensitive
- Sellers who **adopt a new feature** are more sophisticated
- Products that **join a program** have owners who care about marketing

> Ignore selection and you get a **biased** estimate: usually too big, because the treatment went to the units that were going to do well anyway.

Three ways to rebuild the control group

	Approach	In one line	What it needs
1	**Regression with controls**	compare treated and untreated **holding the confounders fixed**	you measured **every** confounder
2	**Matching**	pair each treated unit with an untreated **look-alike**	same as 1, plus look-alikes exist
3	**Difference-in-differences**	compare **changes over time** in treated vs. untreated	the two groups would have **moved in parallel** without the treatment

- Approaches 1 and 2 use the **same assumption** and differ in mechanics. Approach 3 uses a different one, and needs data **before and after**
- Tuesday's exercise: approaches 1 and 2, by you. Today's case: approach 3, on real Amazon data

Approach 1: Regression with Controls

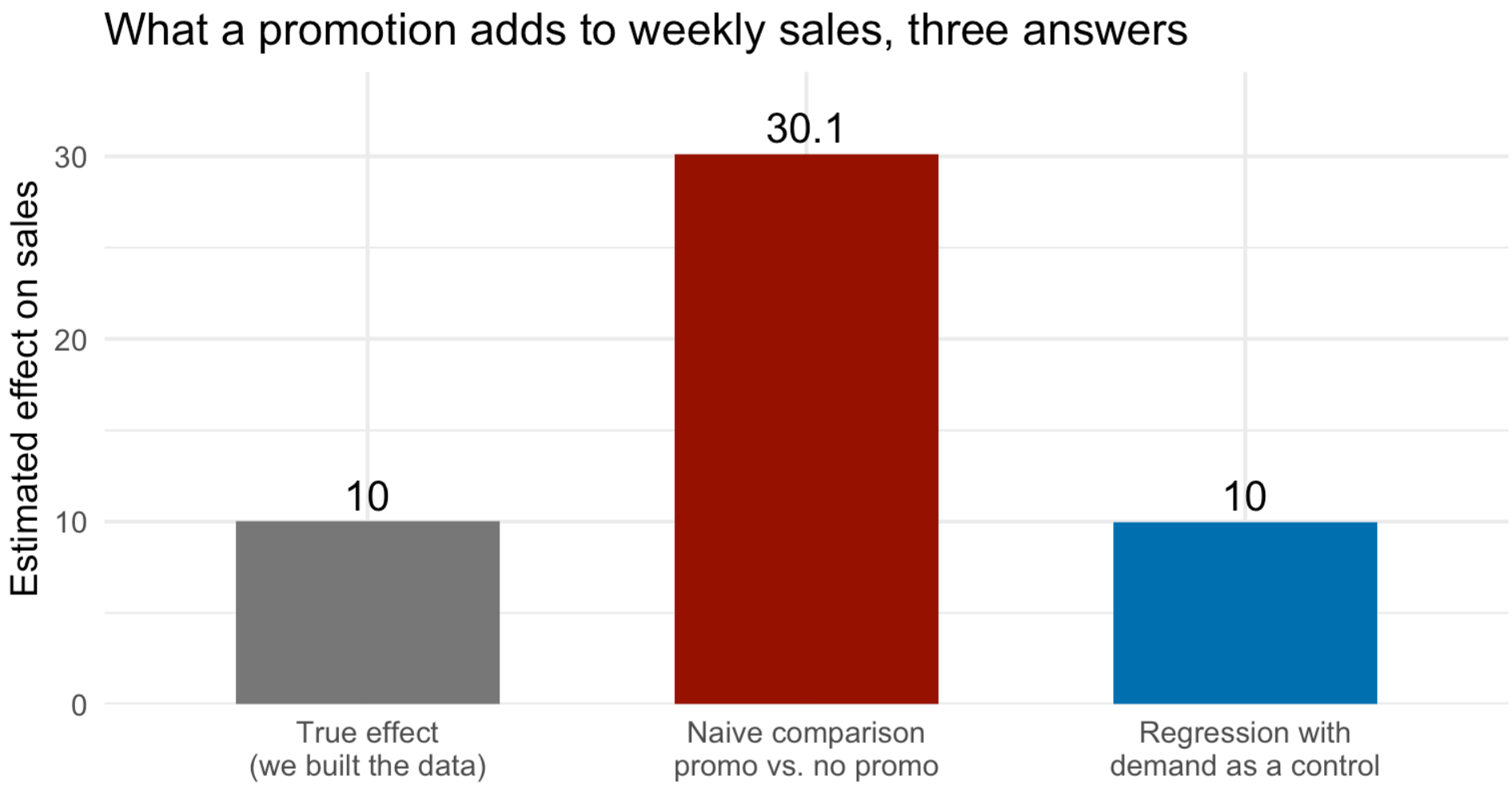
Regression with controls: the idea

Compare treated and untreated units that are **similar on the confounders**, by putting the confounders in the regression:

$$\text{Sales}_i = \alpha + \tau\,\text{Promo}_i + \beta\,\text{Demand}_i + \varepsilon_i$$

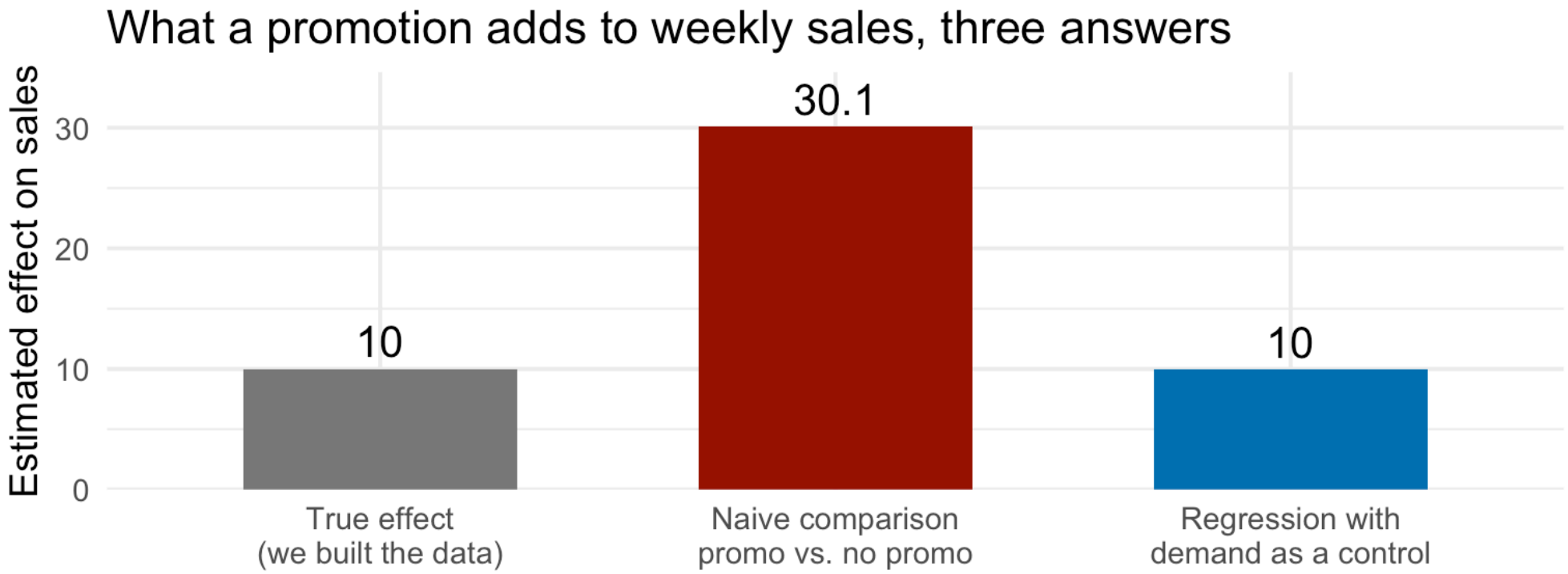
- τ is the difference in sales between promo and no-promo stores **with the same demand**: the regression from week 4, used for a new purpose
- **Key assumption**: you measured **every** confounder and put it in. Then, among stores with the same demand, who runs a promotion is as good as random
- **Risk**: one confounder you did not measure, and τ is biased again. The simplest method, and the weakest

Does the promotion work? Three answers



400 simulated stores, 42% run a promotion. We built the data, so the true effect is **10**. What do you see?

Does the promotion work? Three answers, explained



- **Naive comparison**: promo stores sell 30.1 more. Far too big: in these data the promo stores' demand index is 0.98 higher, so most of the gap is demand, not the promotion
- **Regression with demand as a control**: 10, right on the truth. Demand was the only confounder, and we had it in the data
- The difference between 30.1 and 10 is the **bias** of the naive comparison. The naive number would have a manager tripling the promotion budget

When you do not have the confounder

A retailer sends coupons to customers its model **predicts will be big spenders**. You have past purchases, demographics, and category preferences, so you run:

$$\text{Sales}_i = \alpha + \tau\,\text{Coupon}_i + \beta\,X_i + \varepsilon_i$$

- The problem: the coupon went to customers because of their **predicted future demand**. Your controls capture part of that, not all of it
- The part they miss is in ε_i, correlated with the coupon. τ is biased, and **no amount of data on the same variables** fixes it
- There is no test that tells you whether you have all the confounders. You argue it, and a skeptic argues back

Approach 2: Matching

Matching: the idea

Instead of adjusting with a formula, **build the control group by hand**: for every treated unit, find an untreated unit that looked the same **before** treatment.

1. **Estimate the propensity score**: the probability of being treated, given the observables (a logit of treatment on the confounders, the logit from week 4)
2. **Match**: pair each treated unit with the untreated unit whose score is closest (nearest neighbor)
3. **Compare** the outcomes within the pairs

- Good when: you have **many** observables that explain who got treated, and enough untreated units to find look-alikes
- Same assumption as regression: **selection on observables**. Matching cannot fix a confounder you did not measure

Matching by hand: ten customers

Four customers got the coupon, six did not. The propensity score is each customer's probability of getting the coupon, from their observables.

Customer	Group	Propensity score	Sales
T1	coupon	0.80	70
T2	coupon	0.62	58
T3	coupon	0.45	50
T4	coupon	0.30	44
C1	no coupon	0.81	66
C2	no coupon	0.60	55
C3	no coupon	0.47	46
C4	no coupon	0.28	43
C5	no coupon	0.10	30
C6	no coupon	0.05	25

Which no-coupon customer would you pair with each coupon customer? And which no-coupon customers have no look-alike?

Matching by hand, explained

Treated	Treated score	Matched control	Control score	Treated sales	Control sales	Difference
T1	0.80	C1	0.81	70	66	4
T2	0.62	C2	0.60	58	55	3
T3	0.45	C3	0.47	50	46	4
T4	0.30	C4	0.28	44	43	1

- Each treated customer gets the control with the **nearest score**. Within the pairs, the coupon adds 4, 3, 4, 1: average **3.0**
- C5 and C6 have scores far from any treated customer: **no look-alike**, so they are dropped. That is matching's safeguard against comparing unlike units
- The naive comparison, all four treated vs. all six controls, says 11.3: the two unmatched controls are low spenders and drag the control average down
- The matched estimate is the **ATT**: the effect **on the treated**, the customers who actually got the coupon

Matching in practice

- **Caliper**: the maximum score distance you accept for a pair. Tight caliper, better matches, fewer of them. Loose caliper, more pairs, worse matches. You will try both on Tuesday
- **Check balance after matching**: the matched treated and control groups should have the same average observables, like the balance tables from Tuesday. If they do not, the matching failed
- **Common support**: if the treated units have no look-alikes at all (every treated score is above every control score), matching cannot help, and neither can regression
- In R: `matchit()` from the `MatchIt` package does steps 1 and 2; a regression on the matched data does step 3

Approach 3: Difference-in-Differences

A policy question

City A **bans** short-term rentals (Airbnb); City B does not. We observe average house prices in both cities, **before and after** the ban.

City	Avg. price before	Avg. price after	After minus before
City A (bans Airbnb)	$500k	$520k	+$20k
City B (no ban)	$400k	$430k	+$30k

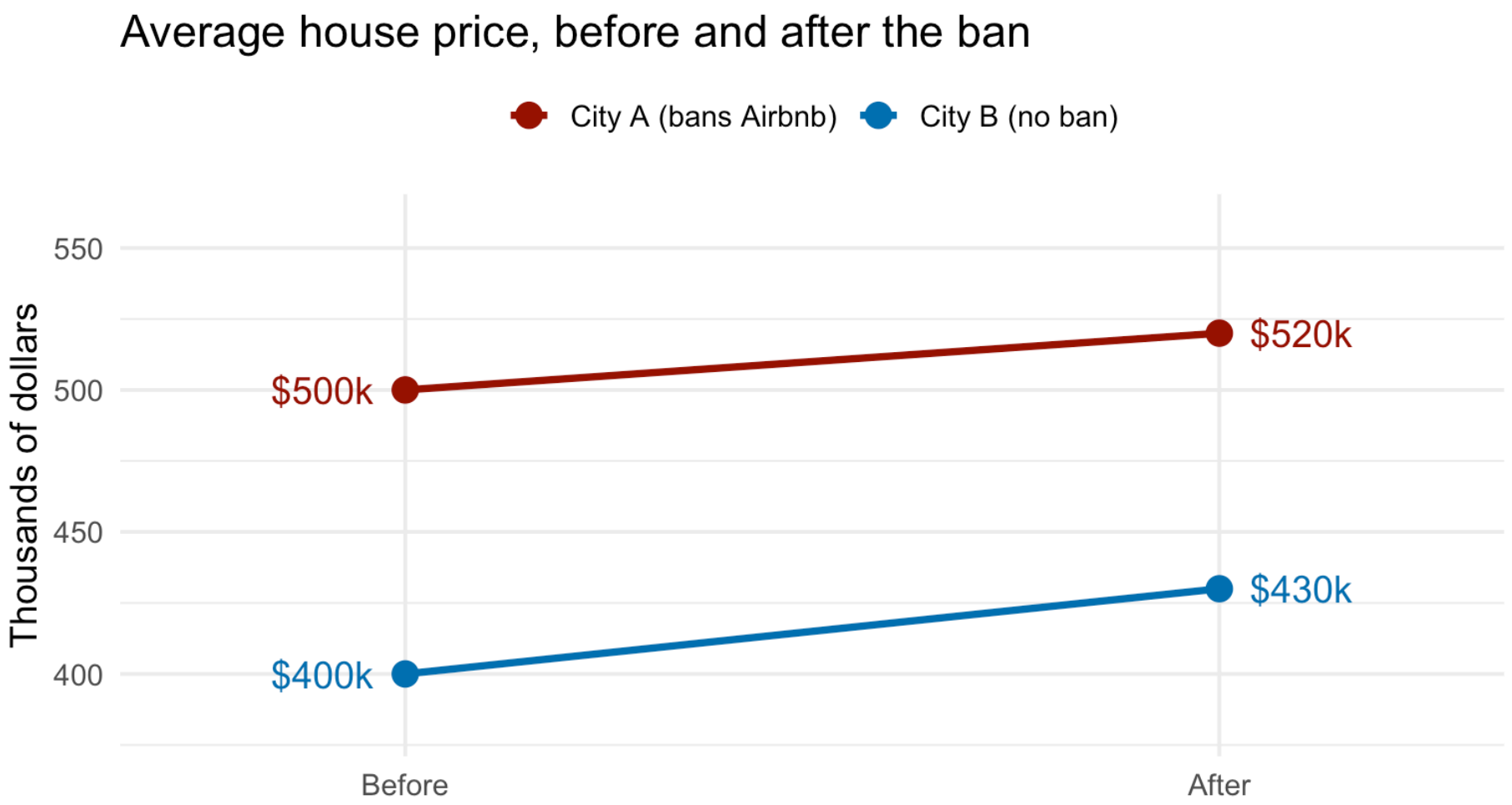
What did the ban do to house prices in City A?

A policy question, explained

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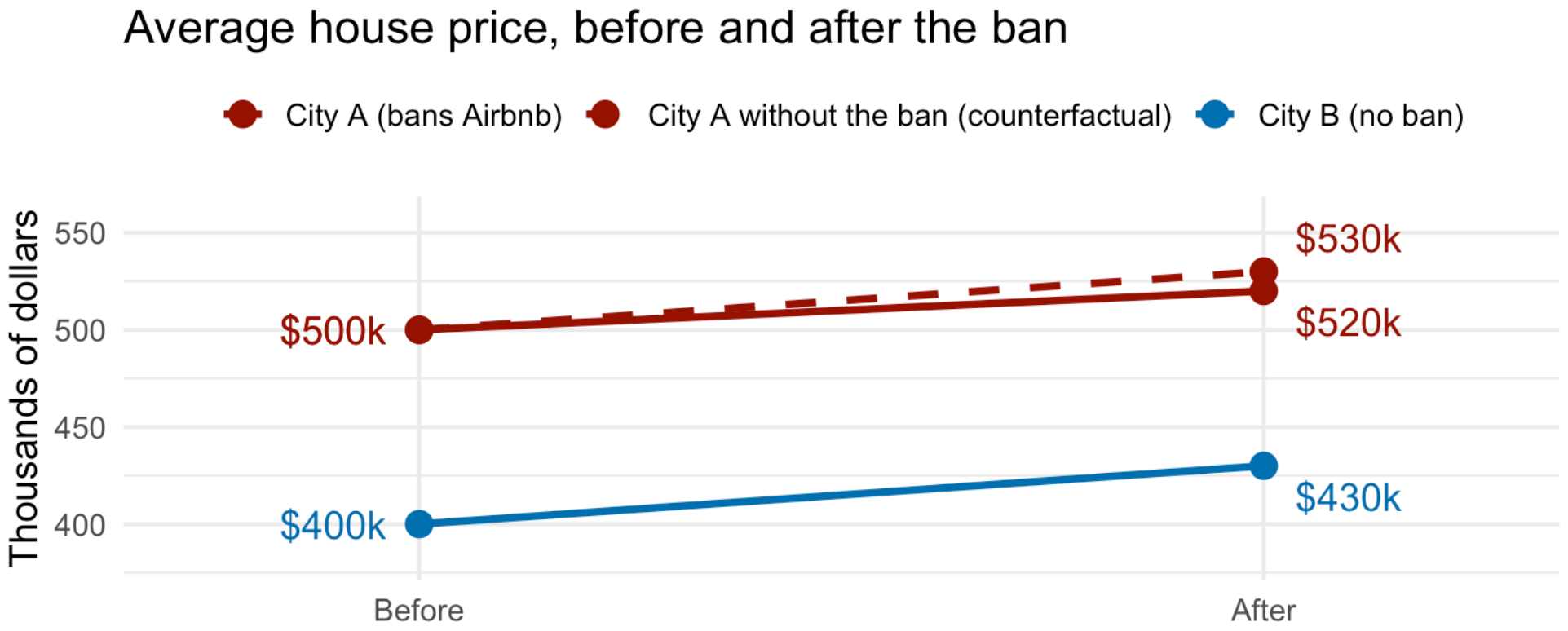
- **+$20k** (before vs. after in A) is wrong: prices rose **everywhere**, by $30k in City B, which had no ban. Time is a confounder
- **+$120k** (A vs. B after) is wrong: A was already $100k more expensive **before** the ban. The cities are different kinds of places
- **Difference-in-differences** removes both: the change in A minus the change in B, $20k − $30k = **−$10k**. The ban lowered prices by $10k relative to what they would have been

The same thing, as a picture



What do you see? Where would City A's line have gone **without** the ban?

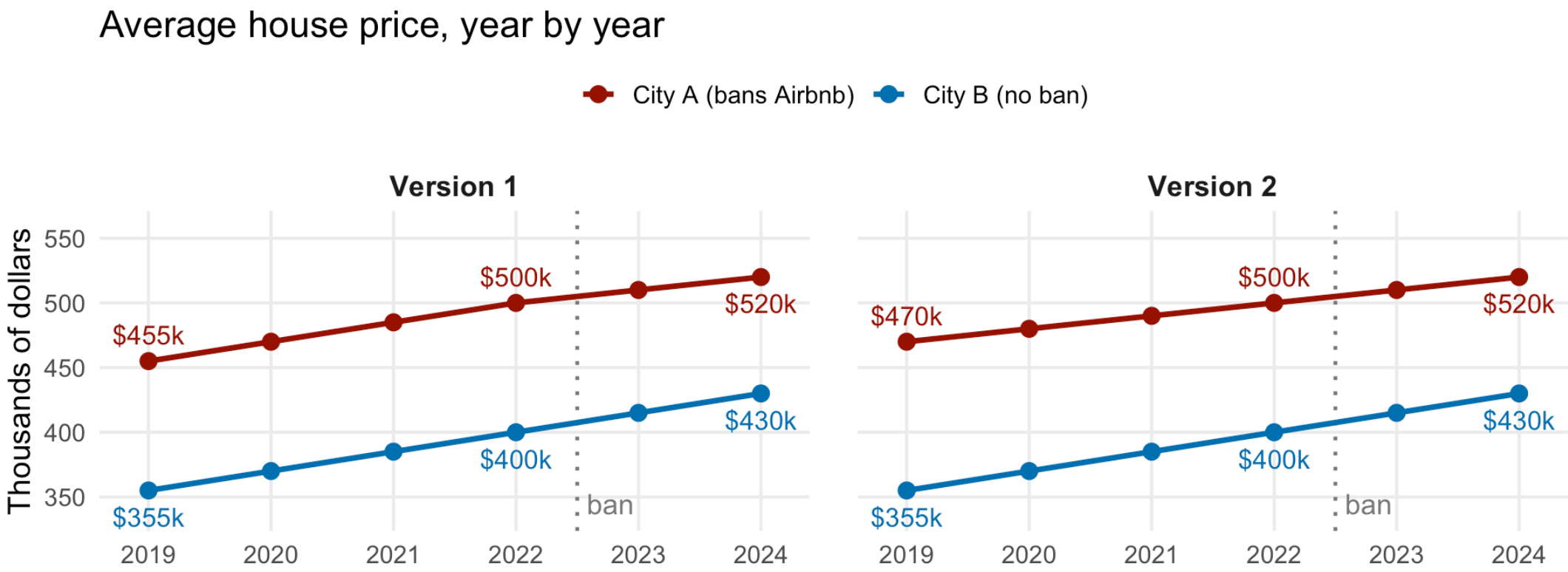
The same thing, as a picture, explained



- The dashed line is the **counterfactual**: City A's starting point plus City B's trend. It is the missing world from Tuesday, rebuilt from City B
- The effect is the gap between where A ended up and where the dashed line says it would have been: $520k − $530k = **−$10k**
- City B's job is to supply the **trend**. City A supplies its own starting level

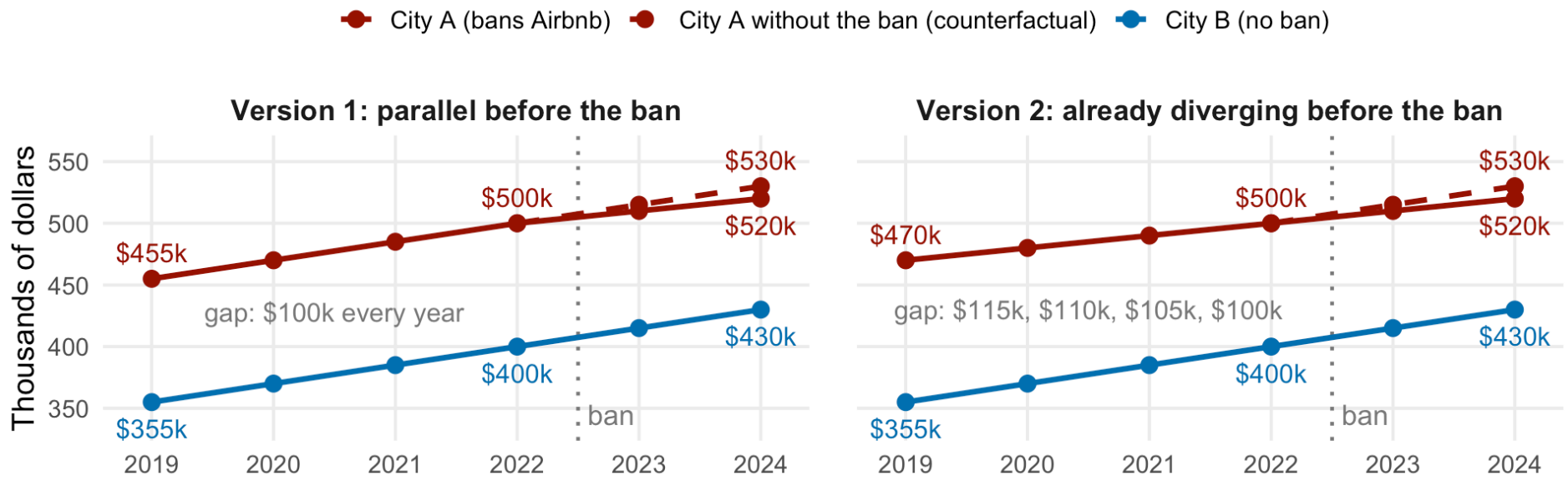
The key assumption: parallel trends

The dashed line borrowed City B's trend. That only works if, without the ban, the two cities would have moved **the same way**: the **parallel trends** assumption. Here are the same two cities **year by year**, in two versions of the world. In both, 2022 (before) and 2024 (after) are the numbers from our table, so DiD says −$10k in both.



What do you see before the ban? In which version do you believe the −$10k?

Parallel trends, explained



- **Version 1**: the gap is $100k every year before the ban and changes only after it. City A would plausibly have kept following City B (the dashed line). The −$10k is credible
- **Version 2**: City A was already growing more slowly, $10k a year against City B's $15k. After the ban it kept its own pace: the ban did nothing, and DiD still says −$10k

> Parallel trends is about a world that did not happen, so it cannot be tested directly. What you can check is the **pre-period**. That is why DiD studies plot the groups period by period, as you will see in the Amazon case.

When DiD fits

Three things need to be true:

1. The treatment happens at a **clear point in time**: a policy change, a product launch, a feature rollout
2. **Some units are affected and others are not**, and the others are a sensible comparison
3. You observe the outcome for **both groups, before and after**

Question	Treated	Control
Does a short-term-rental ban affect house prices?	cities that ban Airbnb	cities that do not
Does a sustainability badge raise sales?	products that get the badge	products that do not
Does free shipping raise sales?	regions where it was introduced	other regions
Does replying to reviews improve future ratings?	restaurants that start replying	restaurants that do not

DiD as a regression

With two groups and two periods, the DiD number is one coefficient:

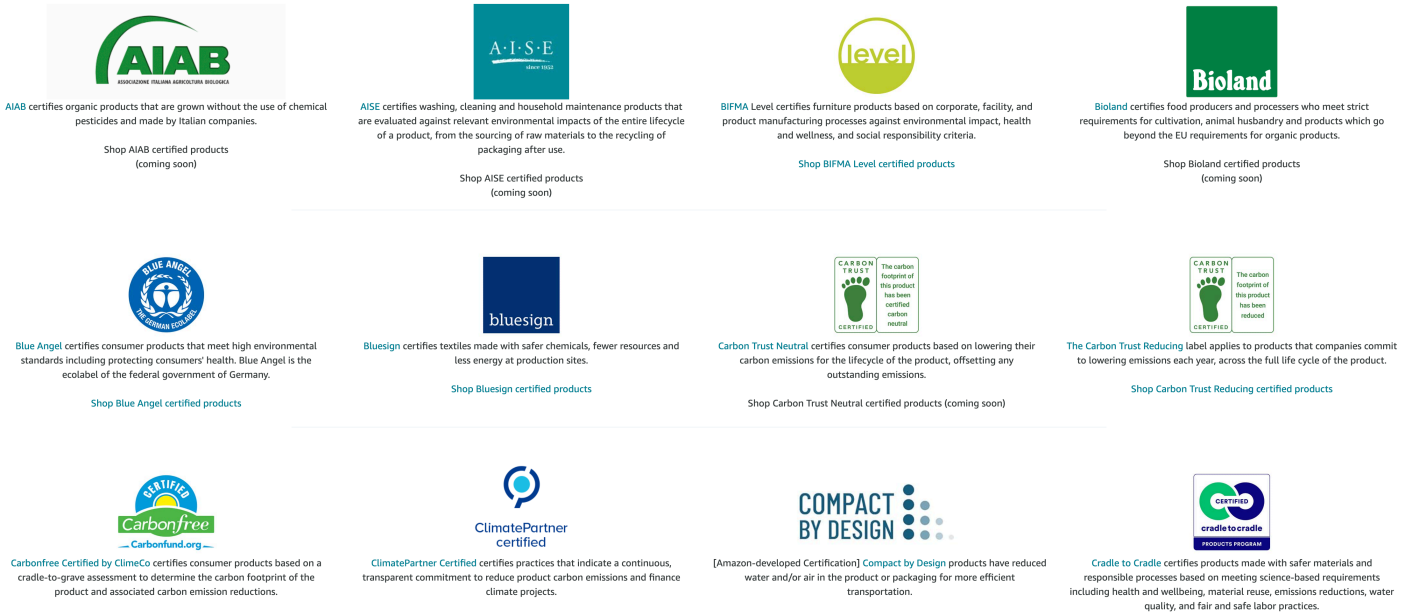
$$Y_{it} = \alpha + \beta\,\text{Treated}_i + \gamma\,\text{After}_t + \tau\,(\text{Treated}_i \times \text{After}_t) + \varepsilon_{it}$$

- β: the pre-existing gap between the groups (A is \$100k more expensive)
- γ: the common trend (prices rose \$30k everywhere)
- τ: the **difference-in-differences**, the effect (−\$10k)

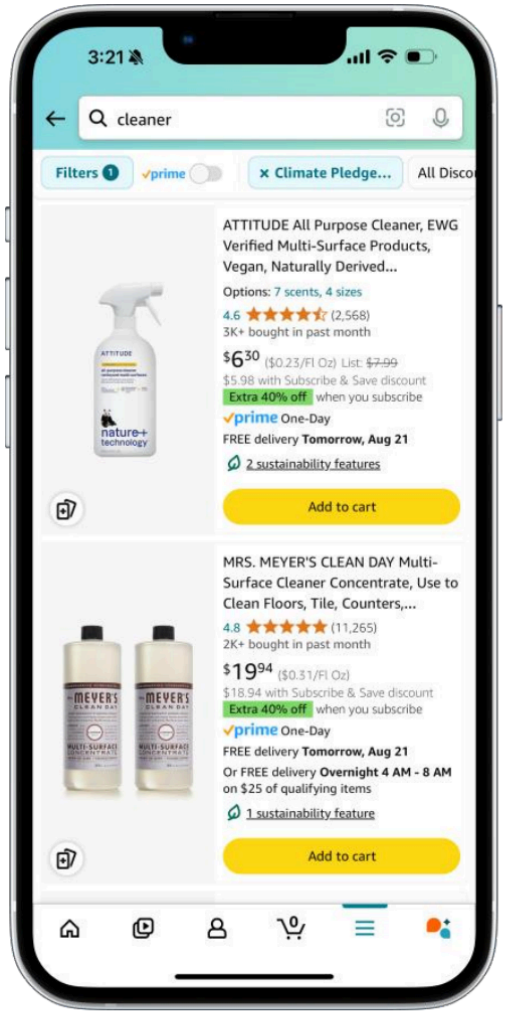
Case: Amazon’s Climate Pledge Friendly

What Climate Pledge Friendly (CPF) is

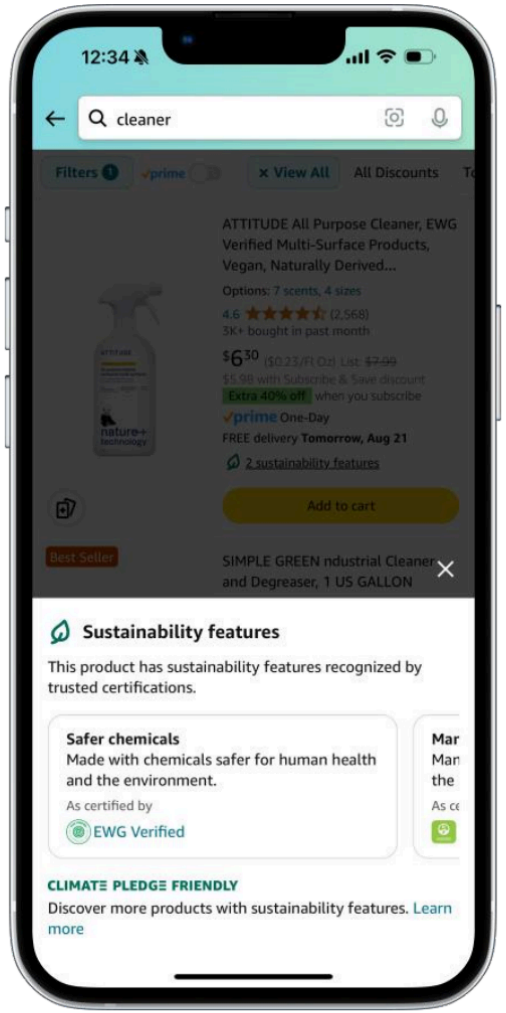
- An Amazon program that helps customers **discover products with sustainability features**
- A product qualifies through one of **60+ trusted certifications** (Blue Angel, Carbon Trust, Cradle to Cradle, …) or Amazon's own
- 14 countries, **2.2 million products**, 271,000 selling partners
- Customers can find CPF products through a **badge** on the product page, a **search filter**, **recommendations**, or a **dedicated storefront**



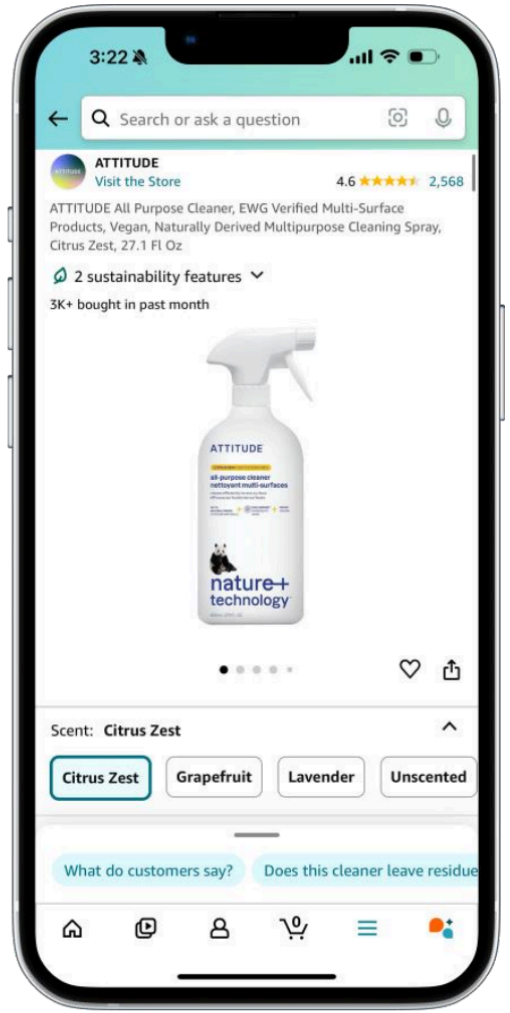
The badge, from search to product page



Search



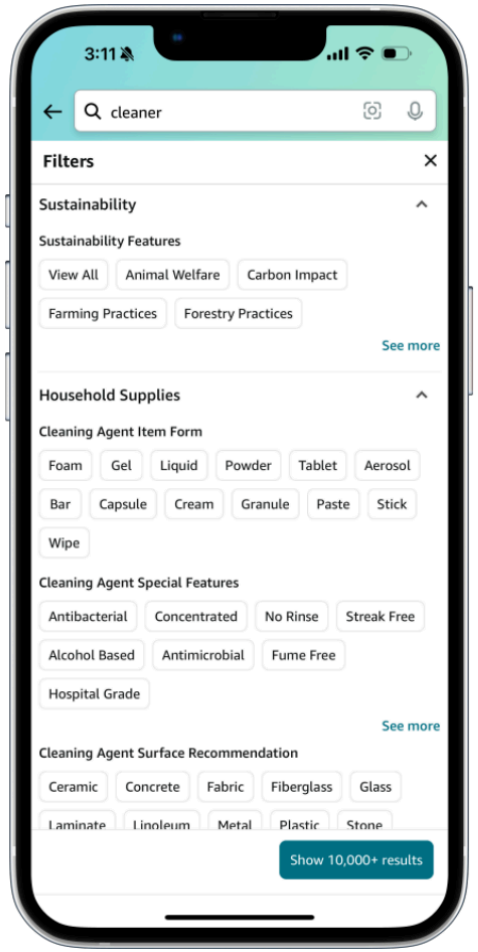
Click



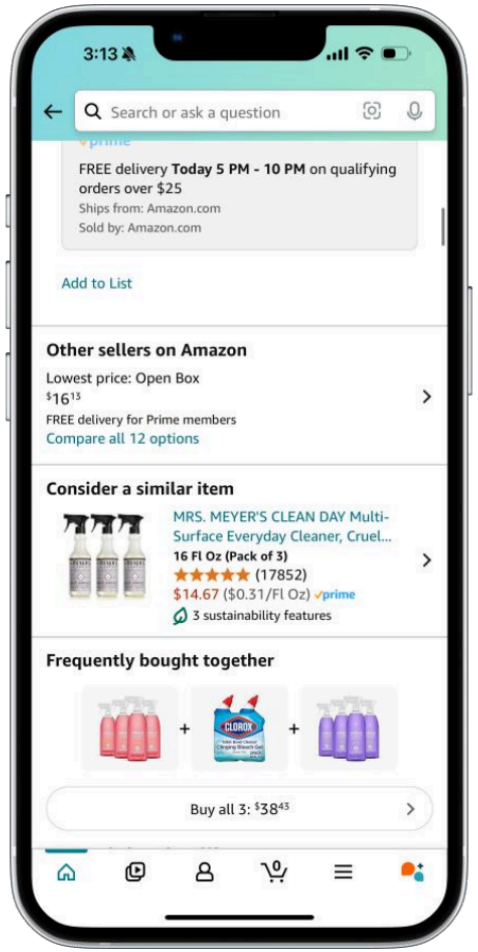
Product page

Three more ways to find CPF products

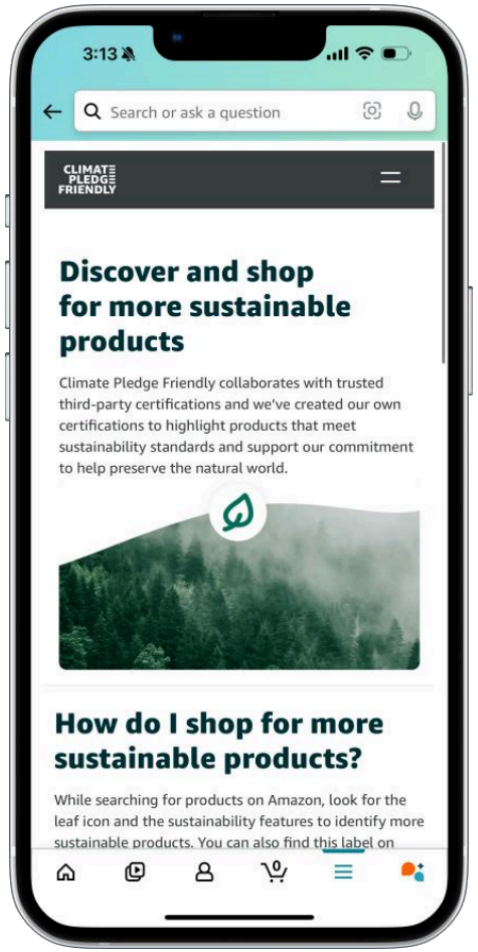
Filter



Reccs.



Dedicated store



The value-action gap

- In surveys, consumers say they **prioritize sustainability**. Whether that shows up in **actual purchases** is an open debate: the gap between stated and revealed preferences
- Whether a program like CPF turns the stated preference into sales depends on
  - **comprehension and trust**: do customers understand the badge and believe it?
  - **beliefs**: cultural and political attitudes toward sustainability
  - **willingness to pay**: is the customer ready to pay more, or anything at all, for it?

> Amazon's question: **does the CPF badge causally change what customers buy?**

Why not just compare CPF and non-CPF products?

CPF products sell more than non-CPF products. Is that the badge?

Using today's vocabulary: what is the confounding story, and what is the selection story?

Why not just compare, explained

- **Confounding**: CPF and non-CPF products differ in ways we cannot see (brand strength, quality, the kind of seller behind them), and those differences move sales on their own
- **Selection**: a product that goes through certification has a seller who **does other things too** (marketing, pricing) around the same time. The sales lift may be theirs, not the badge's

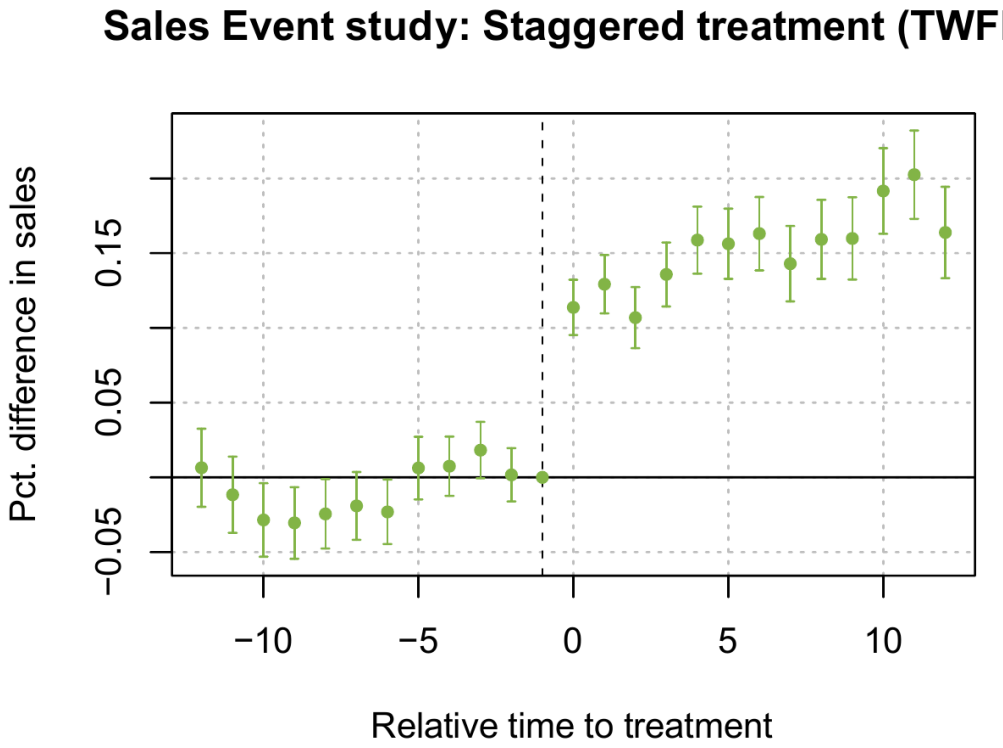
The approach

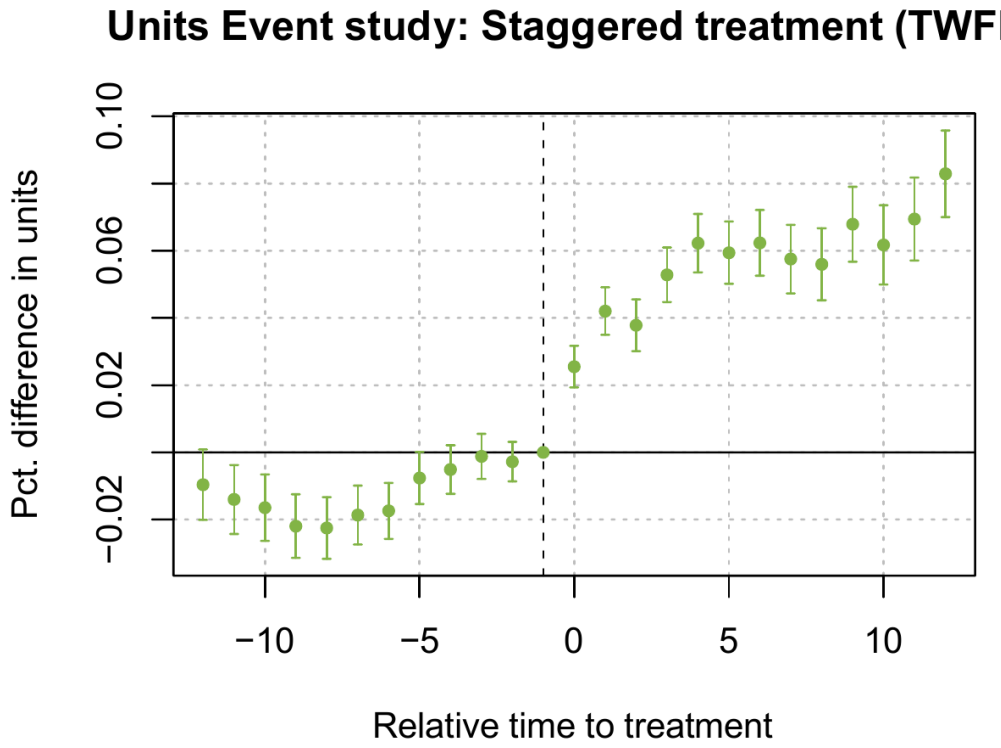
- A **difference-in-differences** that exploits products joining CPF **at different times** (staggered adoption)
- A **short window**, 12 weeks before and after joining, so that fewer other things can change in the meantime
- **Controls** for the things that do change: price, discounts, ad spend, ratings and reviews

The data

- A random sample of products from two populations: those that **eventually join CPF**, and those that **never do**
- About **45,000 products** (36,000 eventually CPF) across three categories: consumables, hardlines, softlines
- **Weekly** data, July 2021 to January 2024: about **4 million** product-week observations
- Two outcomes: weekly **sales** and **units shipped**
- Controls: price, discount, ad spend, ratings, and reviews

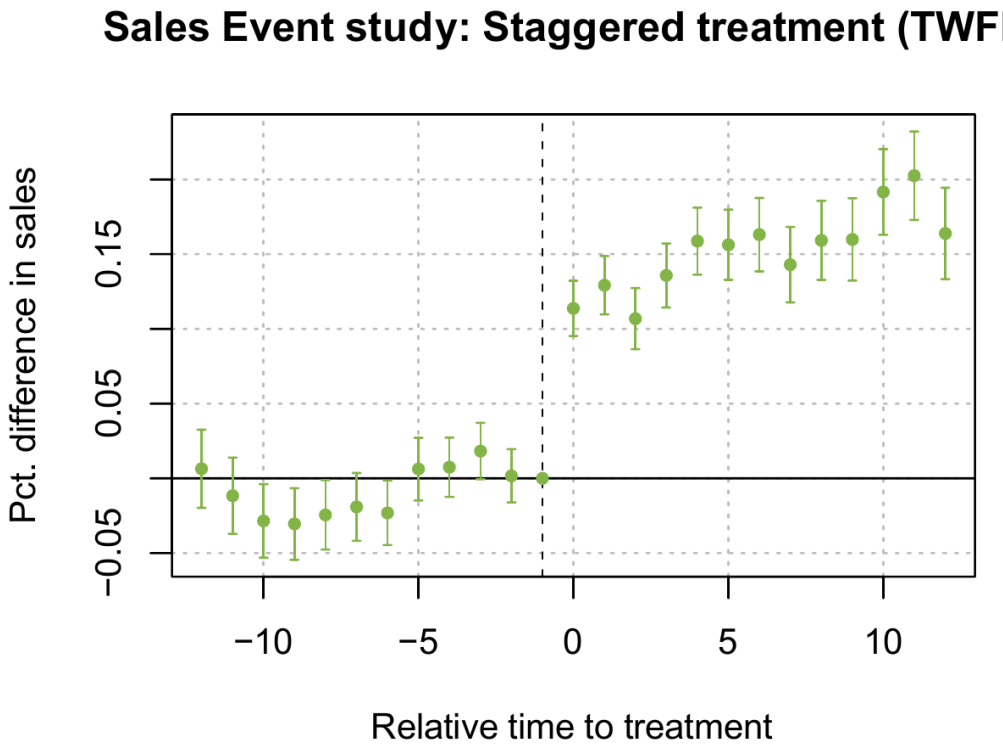
Results: the gap between CPF and control products, week by week

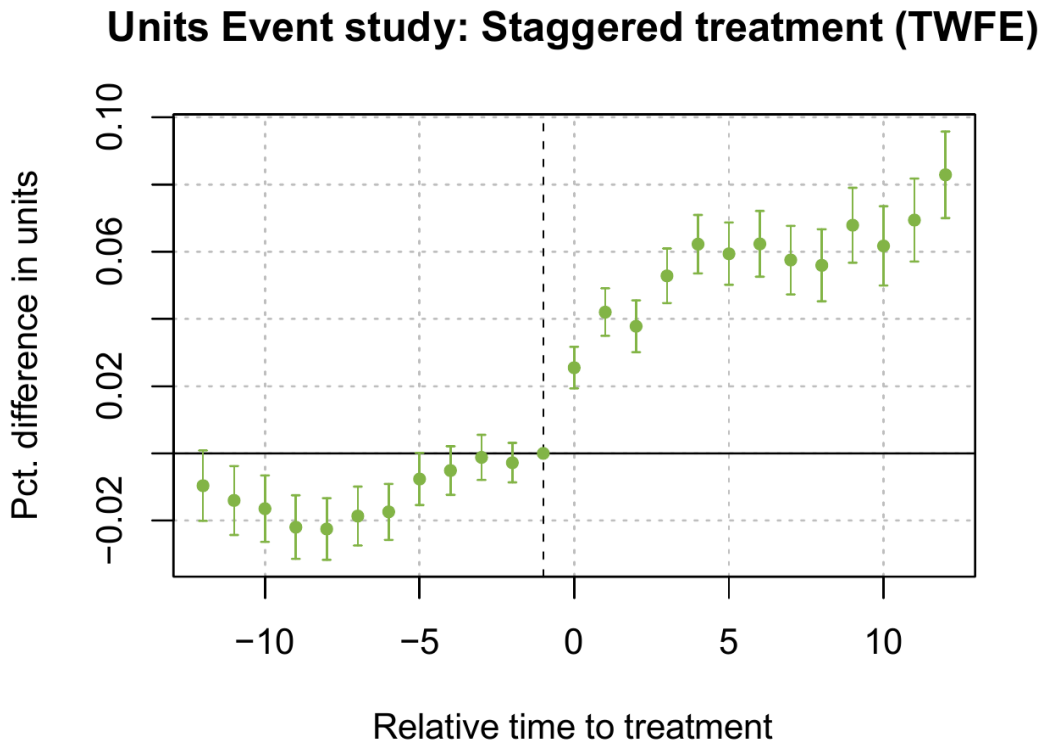




Each dot: the difference between products that joined CPF and the controls, by week relative to joining (week 0). What do you see before and after week 0?

Results, explained





- **Before week 0** the dots sit near zero: CPF products and controls were moving in parallel. That is the pre-trends check, and it passes
- **At week 0** sales jump by about 0.12 to 0.15 log points (roughly **13% to 16%**) and stay there; units jump by about 0.03 to 0.06 (roughly **3% to 6%**) and keep growing
- The jump is immediate, which fits a **badge** that changes what customers see the day it appears

Main results: the numbers

Staggered DiD, product and week fixed effects, standard errors clustered by product. Coefficients are in log points.

	log Sales	log Units	log Sales	log Units
After joining CPF	0.146***	0.059***	0.118***	0.043***
	(0.009)	(0.004)	(0.008)	(0.003)
Controls	No	No	Yes	Yes
Observations	1,365,441	1,365,441	1,242,984	1,242,984

- Without controls: 15.7% more sales, 6.1% more units (a log coefficient b is about $e^b - 1$ in percent)
- With controls for price, promotions, ad spend, ratings, and reviews: **12.5%** more sales, **4.4%** more units
- The controls cut the estimate by about **20%** for sales and **27%** for units: part of the raw jump was sellers doing other things when they joined. The rest is the badge

What the case concludes

- The CPF badge **does** change purchase behavior: with controls, **12.5% more sales** and **4.4% more units** in the 12 weeks after joining
- Making sustainable products **easier to find** lets customers act on the preferences they state in surveys
- Brands that join sustainability programs benefit

> The causal claim rests on **parallel pre-trends** (checked in the plot) and **controls** for what sellers change when they join.

Wrapping Up

Takeaways

- Observational data **can** support causal claims, but every method rests on an assumption about **why some units were treated**
- The enemy is **selection**, or **confounding**: the treated units were different before the treatment did anything
- **Regression with controls** and **matching** assume you measured every confounder. **Difference-in-differences** assumes parallel trends instead, and needs before-and-after data
- The methods differ in **how they rebuild the counterfactual**: adjust with a formula, find look-alikes, or borrow a trend from a control group
- Always **show evidence** for the assumption: balance tables for matching, pre-trend plots for DiD. A causal estimate without it is a correlation with a confident name

Tuesday: the coupon, in three worlds

An in-class exercise, no code given, with your AI assistant:

- One retailer, one coupon, **three versions of the world**: assigned at random, targeted on things you can see, targeted on things you cannot
- You run the naive comparison, regression with controls, and matching in each world, and see **which method survives where**
- The data are simulated, so the true effect is known: you grade your own estimates